



Automatic Greenhouse

Embedded Software for the IOT Project
2025/26

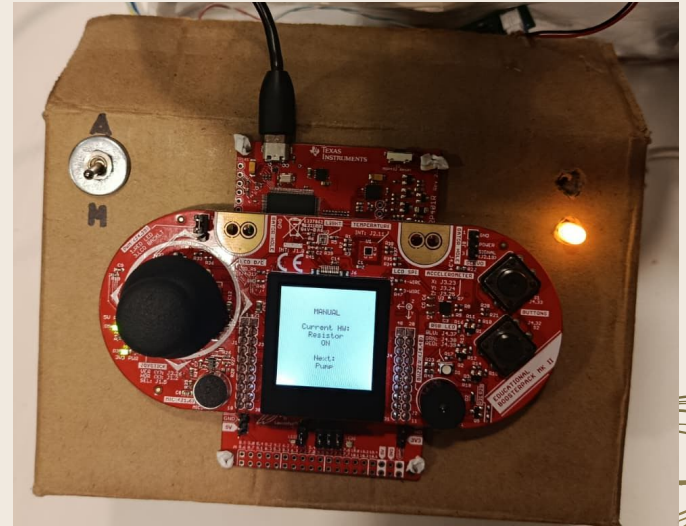


What is it?

Our goal is to build an automated greenhouse system controlled by an MSP432 microcontroller and various peripheral modules. You can find the GitHub repository [here](#) and a video presentation [here](#).

Key functions include:

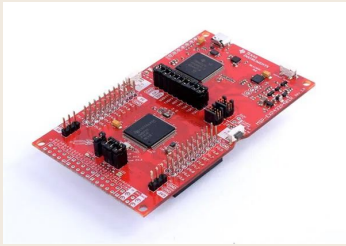
- Automatic watering;
- Temperature stabilization (with a tolerance of $\pm 0.5^{\circ}\text{C}$);
- Humidity control at a fixed percentage;
- Display all the data on an LCD screen.





Components list

MSP-432



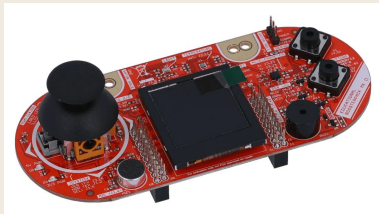
DIY Greenhouse



Air
humidifier



BoosterPack MKII

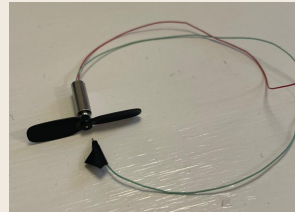


DHT22



(external
humidity and
temperature
sensor)

Fan



Water pump
CJWP08



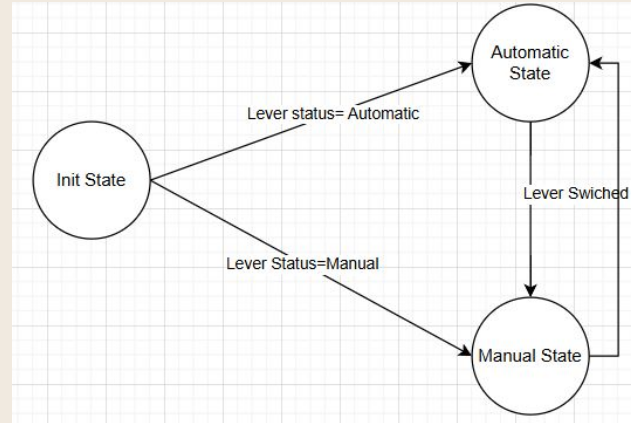


Software Logic

The Greenhouse is implemented by a state machine that have 3 main states:

- Init (starting State)
- Automatic
- Manual

The initial state is always the Init state
switch init all the hardware, after that
the state become Automatic or Manual,
depending on the lever position. Also
you can switch from Automatic to
Manual and vice versa by switching the
lever



```
1 StateMachine_t fsm[] = {
2     {STATE_INIT, fn_INIT},
3     {STATE_MANUAL, fn_MANUAL},
4     {STATE_AUTOMATIC, fn_AUTOMATIC},
5     {STATE_SET_WATER, fn_SET_WATER},
6     {STATE_SET_HUM, fn_SET_HUMIDITY},
7     {STATE_SET_TEMP, fn_SET_TEMP}
8 };
```

